

Testing a Shared Temporal Scale Across Bodily Judgments

Exact cross-task constraints and a secondary analysis of parietal stimulation

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10 October 2026 · CTD-PAPER-v0.1.0

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Status: Complete working manuscript for scientific review; not externally peer reviewed. This is a retrospective secondary analysis of public human data and a mathematical methods application. No new participants, neural interventions, or hardware experiments were conducted. This edition has no assigned DOI. The source experiment, its original findings, and its released fitted parameters belong to D’Angelo and colleagues [1].

Abstract

A physical intervention can affect two bodily judgments without demonstrating that their changes are proportional expressions of one shared temporal process. Conversely, failure of proportionality need not imply separate processes when task-specific measurement relations are nonlinear. We distinguish these claims using two explicit models of positive psychometric widths: a shared multiplicative scale and a shared scalar with stable increasing task maps. For two tasks, the exact smallest uniform log adjustment required by the multiplicative model is one quarter of the range of the between-task log-width differences. The corresponding distance to a common nondecreasing order is obtained by a finite isotonic construction. These are applications of established measurement and approximation principles, with complete proofs and computational checks. We apply them to the 30-participant parietal-stimulation experiment of D’Angelo et al. (2026), retaining the original authors’ prior stimulation, covariance, and Bayesian-model results as inherited evidence. In 180 released width estimates, the mean difference between the tasks’ 8-Hz/13-Hz log gains is -0.0146 (95% CI -0.0933 to 0.0642 ; paired $t_{29} = -0.378$, $p = .708$). The median descriptive adjustment budget for individual proportionality is 4.20%; 25 of 30 estimated profiles have matching condition order. These point-estimate diagnostics are not individual violation tests. A separately specified binomial response model tests all 60 within-person restrictions in the 1,260 count cells; its likelihood-ratio statistic is 73.813, with conditional parametric-bootstrap $p = .105$. Its uncertainty remains separate from the released widths. The results do not identify a neural mediator, establish an experiential scale, or select a global consciousness theory. They supply a reproducible test of explicit model-and-measurement conjunctions and identify the additional calibration needed for a discriminating experiment.

Keywords: body ownership; simultaneity judgment; causal intervention; psychometric width; state-trace analysis; measurement invariance; UCT.

1. From source identification to a discriminating prediction

Suppose two systems behave alike and predict the same consequences. Their successful outputs do not determine which sources their internal consumers use. *Matched Behavior and Source Use* formalized this problem in finite sensorimotor models, giving exact intervention requirements and counterexamples to common identification shortcuts [2]. Its question was organizational: which route is used, under which assumptions? A further question concerns explanation: once an intervention has been performed, what result could distinguish competing accounts of the relevant bodily judgment?

That second question requires more than a change in a measured outcome. An intervention on a shared brain region may influence ownership judgments and simultaneity judgments through a common latent variable, several task-specific paths, a change in decision criteria, or a combination. Even if a shared variable is a good explanation, the tasks may express it on different nonlinear scales. A numerical interaction can then be produced or removed by the measurement relation. This problem has a long history in measurement theory, interaction interpretation, and state-trace analysis [3–9,17]. We do not claim to introduce it.

We examine a specific additional restriction in a real intervention dataset. D’Angelo et al. studied parietal alpha frequency, body ownership, and visuotactile simultaneity [1]. Their Experiment 3 manipulated stimulation condition within participants and measured both judgments. The original study already reported effects on both tasks, individual cross-task correlations of intervention effects, and a comparison of Bayesian causal-inference models with changing sensory uncertainty or changing common-cause priors. The present analysis asks a stricter question: **can each participant’s positive task widths be represented by one stimulation-dependent multiplicative factor and a task factor that remains fixed across stimulation?**

The proposed restriction is not asserted to be a necessary prediction of the original study’s Bayesian model. A shared uncertainty parameter can have different nonlinear consequences for two response functions and their fitted widths. Nor does the restriction follow from Unified Consciousness Theory (UCT), whose structural correspondence axiom does not supply an empirical task-to-experience measurement law [10,11]. It is a deliberately added, falsifiable bridge.

The contribution is consequently narrow and inspectable. We formulate the proportional restriction and a weaker ordinal alternative; derive their exact adjustment budgets; retrospectively test the new constraint on released human data; and separate the resulting evidence from claims about neural mediation or experience. The application also illustrates why the existence of a usable dataset does not remove the need to audit the fitted estimator, the unit of analysis, and the theoretical interpretation of the endpoint.

2. Models and targets

2.1 The measured object

Let i index a participant, $t \in \{O, S\}$ the ownership and simultaneity tasks, and f one of the stimulation conditions. Write $W_{itf} > 0$ for the width parameter of a specified task-response function. An estimate is $\widehat{W}_{itf}$. A psychometric width, a sensory-noise parameter in a generative model, a neuronal integration time, a report of ownership, and a selected experiential property are distinct objects.

This separation matters empirically. Yarrow et al. manipulated judgment strategy and showed that the fitted window of subjective synchrony can change with strategy [7]. Width is therefore not automatically a direct sensory or neural clock reading. In this paper, *shared temporal scale* names a mathematical model of positive task widths. Physical and phenomenal interpretations require additional evidence.

The two models below are applied separately within each participant. The task maps remain fixed across the specified stimulation conditions; that invariance is a substantive assumption, not a consequence of the data table.

2.2 A proportional bridge and its ordinal relaxation

The strict multiplicative bridge is

$$\mathcal{M}_\times : \quad W_{itf} = c_{it}\tau_{if}, \quad c_{it} > 0, \quad \tau_{if} > 0.$$

The task factor c_{it} does not depend on stimulation. Under this bridge, changing stimulation changes both task widths by the same ratio for the same person. Multiplicative task calibration cancels in a within-task ratio. An arbitrary additive offset on the original width scale does not cancel.

A weaker model allows nonlinear task maps:

$$\mathcal{M}_\uparrow : \quad W_{itf} = h_{it}(\tau_{if}),$$

where the maps are strictly increasing and fixed across conditions. Its finite-data closure, $\overline{\mathcal{M}}_\uparrow$, permits nondecreasing maps, including task-specific plateaus. Every proportional bridge is a strict monotone bridge, but the converse is false. These are instances of established common-latent-order and state-trace ideas [3–6].

The models make different observable demands. The proportional model requires equal ratios. The strict ordinal model requires the same ordering and matching ties. Allowing condition-dependent task maps without restriction removes both demands. Calling a variable “shared” then supplies little testable content.

Neither a fitted proportional representation nor an ordinal representation establishes that the physical system has only one cause. These models constrain a selected view of the outcomes. Multiple physical implementations can induce the same table, as the prior source-consumer analysis also illustrates [2].

3. Exact model boundaries

We suppress i in this section. Put $\ell_{tf} = \log W_{tf}$ and $d_f = \ell_{Of} - \ell_{Sf}$.

3.1 Characterization and a sharp proportional adjustment

Proposition 1. The following conditions are equivalent:

$$W_{tf} = c_t\tau_f; \quad \ell_{tf} = a_t + b_f; \quad d_f = d_g \text{ for all } f, g;$$

$$W_{Of}W_{Sg} = W_{Og}W_{Sf} \text{ for all } f, g.$$

Proof. Taking logarithms gives an additive model, and subtracting its rows gives the constant $a_O - a_S$. Conversely, if the row difference is the constant d , choose $a_O = d$, $a_S = 0$, and $b_f = \ell_{Sf}$. Exponentiating recovers the positive multiplicative model. The cross-product equalities are the exponentiated row-difference equalities. $\square$

This is the familiar positive rank-one condition and its log-additive form. With two tasks and three conditions there are two independent restrictions per participant. A single 8-Hz/13-Hz contrast is only one necessary restriction.

Theorem 1. The minimum uniform change on the log scale needed to fit the proportional model is

$$\varepsilon_{\times}^* = \inf_{a,b} \max_{t,f} |\ell_{tf} - a_t - b_f| = \frac{\max_f d_f - \min_f d_f}{4}.$$

Proof. If $\ell_{tf} = a_t + b_f + e_{tf}$ and $|e_{tf}| \leq \varepsilon$, then $d_f = (a_O - a_S) + (e_{Of} - e_{Sf})$. Each difference lies within 2ε of the same constant, so its total range is at most 4ε . This proves necessity. For sufficiency, let

$$c = \frac{\max_f d_f + \min_f d_f}{2}, \quad a_O = c/2, \quad a_S = -c/2, \quad b_f = (\ell_{Of} + \ell_{Sf})/2.$$

The two residuals in condition f are $(d_f - c)/2$ and its negative. Their largest magnitude is exactly one quarter of the range of d_f . $\square$

This elementary Chebyshev-approximation result gives an interpretable diagnostic. At the optimal budget, every corrected width can be kept inside

$$e^{-\varepsilon_{\times}^*} \leq W_{tf}^*/W_{tf} \leq e^{\varepsilon_{\times}^*}.$$

The reported percentage $100(e^{\varepsilon_{\times}^*} - 1)$ denotes an upper multiplicative allowance; the corresponding downward allowance is $100(1 - e^{-\varepsilon_{\times}^*})$. It does not mean that every cell must move by that percentage. It is the smallest common maximum allowance that makes some correction feasible.

For estimated widths, this is a distance of point estimates. It is not a measurement-error estimate, a confidence interval, or proof of an additional neural pathway. Sampling error, unstable response criteria, model misspecification, and task-specific effects can all contribute to the same distance.

3.2 What defensible error bounds would add

Suppose the true log widths belong to a simultaneous rectangular set $\ell_{tf} \in [L_{tf}, U_{tf}]$. Define $D_f^- = L_{Of} - U_{Sf}$ and $D_f^+ = U_{Of} - L_{Sf}$. The smallest proportional distance attained by any log-width table in that uncertainty rectangle is

$$\varepsilon_{\times, \text{box}}^* = \frac{[\max_f D_f^- - \min_f D_f^+]_+}{4}.$$

Proof. Each task difference can be selected independently inside $[D_f^-, D_f^+]$. If all these intervals intersect, the selected differences can be equal. Otherwise every selection must span the gap between the largest lower endpoint and smallest upper endpoint. Choosing values within that gap and their respective intervals attains the minimum range. Theorem 1 completes the result. $\square$

This is a deterministic statement. Separate marginal 95% intervals are not automatically a simultaneous 95% rectangle. We do not use this formula to invent uncertainty for the released fitted widths. It specifies exactly what independently defensible uncertainty would be needed to turn a descriptive incompatibility into a robust one.

3.3 The ordinal model and its sharp distance

Proposition 2. Fixed strictly increasing task maps represent the width table if and only if

$$\text{sign}(W_{Of} - W_{Og}) = \text{sign}(W_{Sf} - W_{Sg}) \quad \text{for all } f, g.$$

Proof. Strictly increasing maps preserve each ordering and each tie of their common input. Conversely, partition the conditions into their common tied classes, assign increasing latent values to the classes, and interpolate each task’s corresponding values with a strictly increasing function on the relevant latent span. $\square$

To measure distance to the nondecreasing closure, let π be a permutation of the F conditions. Define

$$\varepsilon(\pi) = \frac{1}{2} \max_{t, j < k} [\ell_{t, \pi_j} - \ell_{t, \pi_k}]_+.$$

Theorem 2. The smallest uniform log correction admitting a common nondecreasing condition order is

$$\varepsilon_{\uparrow}^* = \min_{\pi} \varepsilon(\pi).$$

For three conditions only six orders need consideration.

Proof. For a fixed order, any corrected nondecreasing sequence within ε of the observed row can remove an inversion of at most 2ε . This gives the lower bound. Conversely, if all inversions are at most 2ε , set

$$m_{t, \pi_k} = \max_{j \leq k} (\ell_{t, \pi_j} - \varepsilon).$$

The sequence is nondecreasing, at least $\ell_{t, \pi_k} - \varepsilon$, and at most $\ell_{t, \pi_k} + \varepsilon$. It therefore attains a feasible correction. Minimizing over common orders proves the result. $\square$

This is standard L_{∞} isotonic reasoning applied to a shared ordering [5,6]. For exactly two tasks, an independent equivalent form is half the largest, across oppositely ordered condition pairs, of the smaller absolute task gap. The supplement proves and checks that equivalence; the analysis uses the transparent six-order enumeration.

The closure qualification matters. Log-width rows $(0, 0, 1)$ and $(0, 1, 2)$ have zero distance to a common nondecreasing order but incompatible exact ties for strictly increasing maps. Strict common-order matrices can approach them arbitrarily closely. Thus zero distance to the closure is not always exact strict representability.

Because the proportional class is contained in the ordinal class,

$$0 \leq \varepsilon_{\uparrow}^* \leq \varepsilon_{\times}^*.$$

An ordinal-compatible table can fail proportionality. A reliably positive ordinal distance would require relaxing at least one of the stable increasing-map, common-scalar, endpoint-model, or measurement assumptions. It would not by itself identify which assumption failed.

3.4 Four instructive counterexamples

Nonlinearity without a second scalar. Let $\tau = (1, 2, 4)$, $W_O = \tau$, and $W_S = \tau^2$. The two tasks have the same strict order and one scalar input, but unequal proportional gains. Here $\varepsilon_{\times}^* = \frac{1}{2} \log 2$ and $\varepsilon_{\uparrow}^* = 0$.

An ordinal conflict. For log rows $(0, 1, 2)$ and $(0, 2, 1)$, the last two conditions reverse. Both distances are $1/2$. Moving both last values in both rows to $3/2$ attains the boundary. A smaller uniform correction cannot remove the reversal.

Unrestricted task drift. Take $\tau_f = 1$ in every condition and let $h_{tf}(z) = W_{tf}z$. Every positive table is reproduced, with an increasing map in each condition. The shared-variable statement has lost its cross-condition restriction because the task map can change freely.

Cancellation across people. Participant A has log rows $(0, 1, 0)$ and $(0, 0, 0)$; participant B has the rows interchanged. Both have proportional distance $1/4$, while the mean log rows are identical. A zero population mean interaction cannot establish individual proportionality. Conversely, noise creates positive individual distances even under an exact true model.

The supplied finite verification checks all 729 two-by-three log matrices with entries in $\{-1, 0, 1\}$. Independent feasible constructions and 4,392,225 comparisons against common-order half-grid corrections agree with the formulas. These computations verify implementation and examples; the proofs establish the general finite-condition claims.

4. Data and analysis

4.1 Source and provenance

We use public Experiment 3 data from D’Angelo et al. [1]: 30 analyzed participants, two tasks, three stimulation conditions (8 Hz, sham, 13 Hz), seven stimulus-onset asynchronies, and ten binary judgments in each participant-by-task-by-condition-by-asynchrony cell. The public workbook contains 1,260 yes-count cells, representing 12,600 judgments, and 180 fitted widths. These are repeated observations from 30 people, not 12,600 independent participants.

The seven asynchronies are -400 , -200 , -100 , 0 , 100 , 200 , and 400 ms. Participant row order and the published task/condition column mapping were retained; trial chronology and block identifiers were not available. The source sample was selected for susceptibility to the ownership illusion (30 analyzed from 37 recruited), limiting population generalization. We use the original authors’

pseudorandomized stimulation order and reported procedural controls as source-study assumptions; this reanalysis does not independently validate stimulation targeting or identify a neural mediator.

The OSF file `Experiment_3.xlsx` was downloaded from the authors' linked project. Its SHA-256 is recorded in the reproducibility manifest. The final three fitted-width headers duplicate the ownership labels. Their values were independently matched, cell for cell, to the publisher's Figure 6c source-data sheet, while the first three matched Figure 6a. The count blocks likewise exactly matched Figures 6b and 6d. We preserve original bytes and document the task mapping explicitly, rather than changing the source workbook.

The original authors already reported cross-task correlations of intervention effects, approximately $r = .728$ for sham-minus-8-Hz changes and $r = .547$ for 13-Hz-minus-sham changes. Our code reproduces those values as a mapping check. Their correlations and Bayesian sensory-uncertainty comparison are not new findings of this paper. Correlated absolute changes do not entail equal proportional changes.

4.2 Retrospective plan and released-width estimand

This is a retrospective analysis. The source article and some aggregate findings were inspected before selecting the question. A local analysis plan was saved before computing the new inferential contrasts; it is not a prospective preregistration. The model-distance diagnostics and tolerance sensitivity analyses are explicitly exploratory.

The primary released-width statistic is

$$\widehat{D}_i = \log \frac{\widehat{W}_{i,O,8}}{\widehat{W}_{i,O,13}} - \log \frac{\widehat{W}_{i,S,8}}{\widehat{W}_{i,S,13}}.$$

We report its mean, participant-level standard error, a paired two-sided t test against zero, and a 95% confidence interval. This is a test of the mean contrast in the released fitted estimates. A corresponding statement about true widths would additionally require appropriate estimator properties. It does not test the hypothesis that every participant's true contrast is zero.

We also report both sham-based cross-task contrasts, with Holm adjustment across those two follow-up tests. A participant bootstrap and a signed-rank calculation are robustness descriptions, not independent replications. All six widths from a participant remain together when resampling. Tolerance analyses use explicit margins, including a 1.10 multiplicative margin; these are illustrative choices, without an independent psychophysical or phenomenal justification.

For each person's released six-width table, we calculate both exact distances and compare the condition orders. No supplied confidence intervals quantify the sampling error of those 180 estimates. Consequently we report individual distances descriptively and do not label individuals as having rejected a true shared temporal mechanism.

4.3 A separately specified count model

We attempted to reconstruct the published widths from the public count curves with a plainly specified three-parameter Gaussian least-squares fit. This did not uniformly reproduce the released values, particularly for ownership. The publisher workbook confirms the extracted rows and task

mapping, but the exact reason for the estimator difference remains unresolved. We do not transfer bootstrap errors from our count fit to the original authors’ width estimates.

To assess both within-person proportional restrictions while retaining a direct count likelihood, we separately specify

$$Y_{itfs} \sim \text{Binomial}(10, p_{itf}(s)), \quad p_{itf}(s) = A_{itf} \exp \left[-\frac{1}{2} \left(\frac{s - \mu_{itf}}{w_{itf}} \right)^2 \right].$$

Amplitudes and centers are free for every participant, task, and stimulation condition in both models. Under the restricted model, $\log w_{itf} = a_{it} + b_{if}$ with $b_{i,\text{sham}} = 0$. Under the alternative, all six log widths are free. There are 16 versus 18 parameters per person, hence 60 total restrictions across 30 participants. This is an operational response-shape model, not the original Bayesian causal-inference model or a mechanistic derivation of a neural clock.

Fits use deterministic multiple starts, analytic gradients, nested-model checks, and recorded bounds. The likelihood-ratio statistic is twice the summed difference in best attained negative log likelihoods. A χ^2_{60} reference is reported descriptively. A parametric bootstrap simulates and refits all 30 participants under their fitted restricted models, using 199 replicates and a recorded seed. This calibrates the statistic conditional on this response model and its fixed participant parameters. It does not supply missing trial-order or block-level dependence, establish global optimality, or reproduce a population sampling model.

Conditional binomial independence, the Gaussian response shape, stable cell probabilities, fitting bounds, and numerical convergence are material assumptions. With only aggregate counts, within-session dependence and criterion drift cannot be empirically reconstructed. Bootstrap uncertainty and numerical failures are therefore reported rather than hidden behind the large nominal judgment count.

5. Results

5.1 The released-width mean constraint was not rejected

The mean 8-Hz/13-Hz cross-task log-gain contrast was -0.01456 , with standard deviation 0.21096 and standard error 0.03852 across 30 participants. The 95% interval was $[-0.09334, 0.06421]$, with $t_{29} = -0.37810$ and two-sided $p = .70811$.

Exponentiating gives a geometric ratio of task gains of 0.98554 , with a 95% interval $[0.91089, 1.06632]$. This interval describes uncertainty about the mean log contrast transformed back to a ratio. It is not a range containing every participant’s gain ratio.

The participant-bootstrap mean interval was $[-0.08989, 0.05924]$. The signed-rank result also did not detect a systematic shift ($p = .96767$). These analyses give no compelling evidence against this particular necessary mean restriction. They do not establish exact equality, individual proportionality, or a single neural mechanism.

Contrast of task log gains	Mean	95% CI	Two-sided p
8 Hz versus 13 Hz	−0.01456	[−0.09334, 0.06421]	.70811
8 Hz versus sham	0.01418	[−0.03754, 0.06590]	.57930
13 Hz versus sham	0.02874	[−0.02759, 0.08508]	.30534

Table 1. New contrasts calculated from the released fitted widths. The two sham-based follow-up tests both have Holm-adjusted $p = .61067$. They and the primary contrast are algebraically related; the table is not three independent confirmations.

At the illustrative equivalence margin $\pm \log(1.10)$, the two one-sided test has $p = .02244$, and the 90% interval is $[-0.08001, 0.05088]$. This sensitivity result says that the estimated *mean* contrast can satisfy that numerical equivalence criterion. The margin was not independently established as scientifically negligible, and the test does not justify exact or individual equivalence. It must not be converted into a threshold for experience.

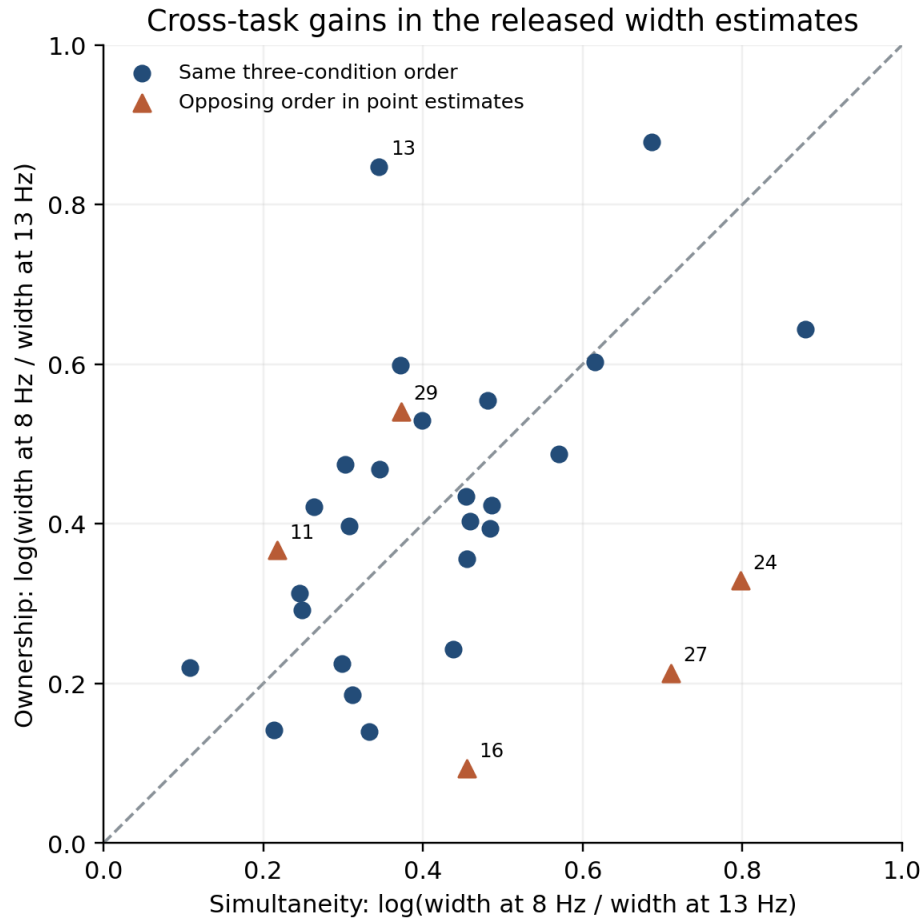


Figure 1: Participant-level log gains calculated from the 180 released width estimates. Each point is one participant; the diagonal denotes equal 8-Hz/13-Hz log gains. Separation from the diagonal is a fitted-estimate contrast, not a calibrated individual violation.

5.2 Individual model distances expose a different question

The median proportional adjustment budget was $\varepsilon_x^* = 0.04112$ log units, corresponding to an upper multiplicative allowance of 4.20%. The interquartile range of that percentage was 1.91%–6.05%, and the largest value was 13.39%. Under an illustrative 10% per-cell allowance, 27 of the 30 released matrices admit a proportional correction.

These are exact descriptive properties of the supplied estimates. A 4.20% median allowance is neither a measured 4.20% neural discrepancy nor evidence that a 4.20% discrepancy is negligible. It states how much adjustment is mathematically sufficient and necessary under the chosen uniform log metric.

Twenty-five participants had exactly the same stimulation ranking across the two released task profiles. For those point estimates the ordinal distance is zero, despite possible nonzero proportional distance. Five had an opposing condition ordering. Without a jointly calibrated error model for the released estimator, these five are not declared true violations of a monotone shared scalar. The contrast between the distances nevertheless makes the assumption hierarchy visible: equal ratios impose a stronger restriction than compatible order.

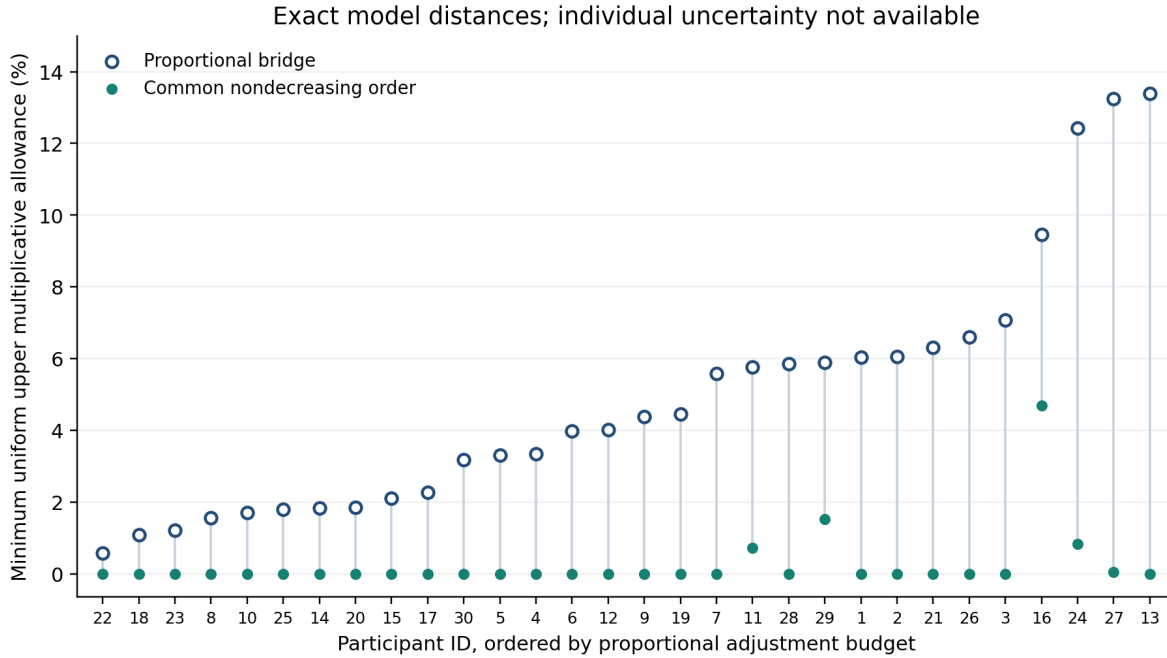


Figure 2: Exact point-estimate adjustment budgets for both model classes. Participants are ordered by the proportional budget. The ordinal budget cannot exceed the proportional budget. Zero ordinal distance means compatibility with the nondecreasing closure; it does not identify a physical mechanism.

5.3 The count likelihood tests the full set of restrictions

For the independently specified binomial Gaussian model, the unrestricted and restricted negative log likelihoods were 4633.20782 and 4670.11455. The likelihood-ratio statistic was 73.81345 for the

60 proportional restrictions. The descriptive χ^2_{60} reference gives $p = .10837$. All original participant fits reported convergence.

The 199-replicate parametric bootstrap yielded 20 statistics at least as large as the observed statistic. The plus-one tail estimate was $p = (20 + 1)/(199 + 1) = .105$. The 95% exact binomial interval for Monte Carlo uncertainty in the simulated tail probability was [.06248, .15095]. This interval concerns finite simulation error, not uncertainty about the model assumptions. All 5,970 bootstrap participant fit pairs reported convergence after ten targeted numerical retries. The restricted and unrestricted fits frequently reached the amplitude upper bound (29 and 30 participants, respectively), supporting the use of the explicitly conditional bootstrap rather than treating the chi-square reference as an exact calibration. The proportional restriction was not rejected at the conventional .05 level under this specified model.

The unrestricted fit's deviance relative to independent saturated count cells was 696.21300. A nominal 720-degree-of-freedom reference gives $p = .73114$, but sparse cell sizes, boundary behavior, and ignored dependence limit this diagnostic. A nonsignificant deviance is not proof that the response shape or latent interpretation is correct.

This analysis addresses a stronger question than a zero average 8-Hz/13-Hz contrast: all 60 within-person proportional restrictions are tested jointly within the stated count model. Its widths are estimates under a separately specified fit and are not assumed interchangeable with the source's released estimates. Neither analysis establishes the physical number of mechanisms, and agreement in their general direction is not an independent replication because both derive from the same study.

5.4 Reproducibility boundary

The source's six-column width table and four publisher source-data blocks were exactly cross-checked. The inherited cross-task intervention correlations were reproduced. The extraction is therefore traceable. The attempted least-squares width reconstruction remains incomplete and is included as a diagnostic, with its fitting choices and examples. No participant matching algorithm was used to relabel people or repair the source.

The unresolved reconstruction prevents attaching trial-derived confidence intervals to the released parameter-level distances. It does not prevent reproducing the stated arithmetic on those released values or fitting an independently declared model to the count cells. The distinction is carried through the code, tables, figures, and claim ledger.

6. What the experiment can distinguish

6.1 Model conjunctions, not theory names

The proper unit of a discriminating prediction is a concrete theory instance together with an intervention model, parameter restrictions, and a measurement bridge. A broad label such as predictive processing or UCT does not supply all of these. Table 2 records what is tested here.

Specified account	Observable restriction	Status in this analysis
Multiplicative width bridge	Equal within-person ratios across conditions	Tested under specified estimators; not established
Scalar with stable increasing maps	Same order and matching strict ties	Descriptive distances; individual uncertainty missing
Scalar with condition-specific maps	No cross-condition restriction	Not discriminable from widths alone
Original Bayesian uncertainty model	Its own nonlinear response predictions	Original comparison inherited; not retested here
UCT C1 without an endpoint bridge	No signed numerical width prediction	Not selected by these data
IIT, GNWT, or HOT as broad families	No task-specific prediction supplied here	Not tested

Table 2. A scope comparison. A failure of the proportional bridge would not automatically count against all common-scalar models, the original Bayesian account, or an entire consciousness theory. A successful fit would likewise not select one of them.

Experimental theory comparison can be more informative when proponents agree in advance on divergent predictions and their observation rules. The Cogitate collaboration illustrates the value and difficulty of such explicit commitments in consciousness research [12]. The present analysis adopts the requirement for explicit commitments; it does not import that study’s predictions into body ownership or assert a comparable adjudication of global theories.

6.2 UCT interpretation

UCT C1 identifies an admitted actual process token’s complete physical organization with its experiential organization, while distinguishing that complete object from finite views and estimates [10]. Its universal-experience commitment does not make basic experience conditional on prediction, report, bodily ownership, intelligence, or the success of a width-model test. Those commitments remain axiomatic in this paper.

The present measurements are finite task outcomes. Neither the 180 widths nor a fitted latent scale determine complete physical organization. Even an independently identified neural route would still require an appropriately justified bridge to a particular experiential endpoint. Agency, ownership, tactile sensitivity, and familiar mineness must not be silently substituted for one another; empirical work already distinguishes aspects of these bodily judgments [13,14].

For UCT, the methodological advance is a way to state an additional bridge sharply enough to fail. It does not establish C1, infer a unique subject from a statistical factor, or exclude nested and overlapping actual processes. A common temporal coordinate across tasks also does not imply that their experiences are identical. Distinct experiential structures can share an organizational feature.

This separation prevents a tempting but invalid inference: a brain perturbation changes an ownership report, therefore a specified internal consumer has been identified, therefore UCT is confirmed. Each implication requires its own evidence. The source experiment supports important claims under its own design; the secondary analysis supplies a narrower test of a new operational restriction.

6.3 Why the result remains useful without a theoretical winner

The data do not compel rejection of the specified proportional response model. The appropriate response is to retain the result and identify what was actually constrained. An uninformative generic claim has been replaced by exact testable equalities, explicit alternatives, descriptive lower bounds on model adjustment, and a count-level likelihood test.

The two distances also prevent the wrong follow-up experiment. If only proportionality fails, the next comparison should address nonlinear or unstable task bridges before asserting two separate neural processes. If common order fails beyond defensible uncertainty, the experiment must distinguish task-map drift, a second latent influence, and endpoint-model misspecification. Repeating a group-mean contrast cannot answer those questions.

7. A concrete prospective extension

The original source-consumer question remains physically stronger than the present dataset: keep selected action and prediction outputs matched while changing which information reaches a verified consumer, then test an independently specified experiential endpoint. No such selective motor-command/proprioceptive intervention was performed in this reanalysis. The following extension specifies the missing work without presenting it as an obtained result.

First, define one actual installation and a limited target: for example, an identified sensorimotor prediction consumer and a separately calibrated ownership or agency judgment. Source identities, consumer ports, timing, task instructions, learning state, and the scope of output matching must be recorded. Equality of a selected action trajectory does not establish equality of all afferent information. Conversely, a decoding signal at a region does not establish consumption by the downstream process.

Second, specify competing models before selecting diagnostic interventions. A fixed match-only account predicts equal endpoint distributions whenever its selected prediction-error input and response mapping are matched. A route-sensitive account must name an additional consumer relation and state a distinct, signed or quantitatively bounded prediction for the same endpoint. A broad forward-model or UCT label is insufficient. Prior agency models already show why a match-only account is an especially restricted comparator [15,16]. It should not stand in for all existing theories.

Third, use separate calibration data to constrain each measurement bridge, then freeze it for held-out interventions. A task factor or monotone map estimated anew in every condition can absorb the sought discrepancy. Use repeated measurements and record trial/block order so the joint uncertainty can be propagated through the actual estimator. The interval corollary gives a concrete decision rule when simultaneous log-width bounds are warranted: an independently justified error rectangle must remain separated from the tested model class. Other endpoints require their own corresponding rule.

Fourth, validate intervention selectivity and the actual consumer relation. A source block that also changes a direct pathway can mimic mediation. An apparent rescue can instead be compensation by a newly learned reader. Prior source-consumer work makes these distinctions explicit [2]; the prospective experiment must establish the relevant exclusion and timing assumptions, rather than infer them from endpoint recovery alone.

Finally, report outcomes at the level their premises support. A held-out discrepancy beyond both estimation error and allowed bridge drift can reject a specified model conjunction. The result

becomes comparative evidence between theories only when their alternative, faithful model instances predict distinguishable observations under the same validated intervention. An unchanged endpoint may reflect a shared feature, compensation, insufficient precision, or an insensitive bridge. It cannot establish absence of experience.

This is a prospective design contract, not a power calculation or an approved human protocol. The present aggregate data do not provide the estimator reliability, trial dependence, or scientifically motivated minimum separation needed to justify a new participant number. Those quantities must be estimated in an appropriate pilot or recovered from the original analysis pipeline before prospective recruitment.

8. Limitations and contribution statement

The mathematical results are applications of classical rank-one, Chebyshev, and isotonic principles. The study does not claim a new branch of mathematics, the first common-latent-order test, or the first neural intervention on bodily perception. The internal and external novelty search found no previous report of this exact proportional-width reanalysis in the checked materials, but cannot establish universal priority.

The empirical contribution is restricted to one selected 30-person dataset, released fitted parameters, and an independently specified response model. The analyses are retrospective, share the same source observations, and have no independent replication. The source sample’s illusion susceptibility, the absence of trial-order records, the unresolved source-estimator reconstruction, and the arbitrary illustrative equivalence margins limit inference.

The selected endpoints are judgments and fitted response characteristics. Their relation to sensory uncertainty, neural timing, ownership experience, or familiar self-related experience requires additional validation. The intervention effect does not alone identify a mediator or a particular information-consumer route. UCT C1 remains an inherited interpretive axiom; global consciousness theories were not adjudicated.

The defensible originality statement is therefore: **we formulate and retrospectively test a specific cross-task proportional-width constraint, report exact distances to both proportional and ordinal model classes, and use the resulting evidence to state which mechanism-to-endpoint inferences remain unlicensed.** This is a methods and secondary-analysis contribution. The prospective scientific advance will require independently calibrated bridges and a physically validated intervention that generates genuinely divergent held-out predictions.

Reproducibility, data rights, and disclosures

The accompanying package contains unchanged public input files, provenance and checksum records, the extraction and analysis code, per-participant tables, independent mathematical checks, fitting diagnostics, figures, a claim-level originality audit, and a Chinese research handoff. The original article and publisher source data are attributed to D’Angelo et al. [1] under the source’s stated terms; original files are not relabeled as newly collected data. The package records observed rights information and separates new code from third-party materials.

Computations use Python, NumPy, SciPy, pandas, and openpyxl; figures use Matplotlib. Exact versions, random seeds, numerical bounds, fit statuses, and reproduction commands are included.

The deterministic width contrasts and mathematical checks can be rerun independently of the longer count bootstrap. Finite checks are not human data; parametric-bootstrap samples are explicitly simulated conditional datasets.

Substantial ChatGPT (OpenAI GPT-6 Astra Pro) assistance was used for retrieval, analysis, derivation, internal adversarial review, drafting, and preparation under human direction. Collaborating AI passes are not independent human peer review. No final human line-by-line verification or external scientific endorsement is claimed. The author of record is responsible for deciding whether and how to release a later public edition. No correspondence with source authors was sent as part of this analysis.

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